IRENA
International Renewable Energy Agency

RENEWABLE POWER GENERATION COSTS IN 2025



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About IRENA

The International Renewable Energy Agency (IRENA) is an intergovernmental organisation that supports countries in their transition to a sustainable energy future and serves as the principal platform for international co-operation, a centre of excellence, and a repository of policy, technology, resource and financial knowledge on renewable energy. IRENA promotes the widespread adoption and sustainable use of all forms of renewable energy, including bioenergy, geothermal, hydropower, ocean, solar and wind energy, in the pursuit of sustainable development, energy access, energy security and low-carbon economic growth and prosperity. **www.irena.org**

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HIGHLIGHTS

After more than a decade of steep declines, in 2025, the levelised cost of electricity (LCOE) from renewables began stabilising. Solar photovoltaic (PV) remained at its 2024 level of USD 44 per megawatt hour (MWh). Wind, however, continued to improve, with onshore wind falling to USD 33/MWh and offshore wind to USD 78/MWh. In contrast, most dispatchable renewable technologies[1] recorded higher costs, with hydropower, geothermal and concentrated solar power rising to USD 62/MWh, USD 89/MWh and USD 115/MWh, respectively. Bioenergy was the exception in this group, with its cost declining to USD 86/MWh.

Since 2010, the cost of solar PV has fallen by 89%, onshore wind by 71%, and offshore wind by 63%. This highlights how renewables are now the cheapest source of new electricity in most markets. In 2025, more than 90% of newly commissioned, utility-scale capacity delivered power at a lower cost than the cheapest, newly-installed fossil-fuel-based alternative.

Between 2024 and 2025, new, natural gas-fired generation increased in cost. A turbine shortage roughly doubled the capital expenditure required for a new, combined-cycle plant to about USD 2 400 per kilowatt in the United States. This also pushed the levelised cost of electricity (LCOE) from gas-fired generation towards USD 100/MWh in higher-gas-cost markets, such as Italy, Germany and Japan. In markets where gas is cheaper, the LCOE remained in the USD 50-60/MWh range. Renewed conflict in the Middle East in early 2026 also suggested higher gas and power prices could persist throughout the year.

China remains the global, low-cost benchmark for wind. In 2025, the LCOE of onshore wind in China stood at USD 27/MWh and offshore wind at USD 49/MWh. This compared with USD 141/MWh for offshore wind in the United States. China is also among the lowest-cost solar markets, at USD 36/MWh, a figure close to that of India, at USD 35/MWh and Brazil, at USD 37/MWh.

IRENA's new cost comparisons for high-reliability hybrid systems show that the firm LCOE of solar-plus-battery systems with a 95% reliability fell from more than USD 100/MWh in 2020 to below USD 85/MWh in 2025 at high-quality sites. Further reductions of around 30% by 2030 and 40% by 2035 are projected.

The installed cost of four-hour utility-scale batteries fell by close to 30% in 2025, to around USD 140 per kilowatt hour (kWh) – around 95% lower than its 2010 level. Costs also fell to below USD 70/kWh in China. These cost reductions are transforming the economics of hybrid renewable systems and strengthening their role in meeting demand reliably.

Falling battery costs are driving the rapid growth of hybrid solar-plus-storage systems. In 2025, around one quarter of all newly commissioned utility-scale solar capacity was paired with battery storage, with the firm LCOE of such systems at high-quality sites falling below USD 85/MWh.

[1] *'Dispatchable' renewables include those technologies that can be adjusted to meet demand when required, namely bioenergy, geothermal, hydropower and concentrated solar power.*

Costs of hybrid systems should keep falling over the next five years, but more slowly and unevenly. The costs of mature technologies, such as solar PV and onshore wind, are plateauing, while batteries and long-duration storage are expected to see further cost reductions as deployment increases. At the same time, the rebound in battery metal prices in early 2026 is a reminder that input cost volatility can slow the downward cost trend.

Emerging technologies are advancing rapidly. Sodium-ion batteries are gaining commercial traction, with rising lithium prices strengthening their business case. The rapid displacement of nickel manganese cobalt chemistry by lithium iron phosphate in recent years suggests that markets can shift quickly once a new technology demonstrates a clear cost and performance advantage.

Next-generation perovskite solar cells are approaching commercial deployment, while long-duration energy storage – including compressed air and flow batteries – is expected to see substantial cost reductions as deployment scales. Ocean energy technologies, such as wave and tidal stream, are nearing industrial deployment. At 2 GW of cumulative deployment, these have projected LCOEs of around USD 120/MWh and USD 140/MWh, respectively.

Cost trajectories will depend on manufacturing dynamics. Clean-technology manufacturing investment has fallen by around half since its 2023 peak. This decline has been driven by China's efforts to curb solar overcapacity and the rollback of the Inflation Reduction Act in the United States. These shifts could affect future supply chains, prices and the pace of deployment.

In addition to technology costs, financing remains a key driver of generation costs. Increasingly, where a project is built matters more than what is built: national macroeconomic conditions explain around 2.3 times as much variation in financing costs as technology, underlining the importance of country risk, policy stability and access to affordable capital.

In 2025, renewables helped avoid an estimated USD 480 billion in fossil fuel costs and about 8.4 gigatonnes of carbon dioxide emissions. This confirms the role of renewables not only as the cheapest new power, but also as a pillar of energy security, economic stability and resilience.

Renewables proved their ability to act as an energy cost shock absorber in 2025. Across three import-exposed Southeast Asian economies – Indonesia, Thailand and the Philippines – the existing renewable fleet avoided around USD 5.7 billion of coal and gas purchases in 2025. Valued at the higher fuel prices seen during the March–May 2026 crisis in the Middle East, those same avoided volumes would have been worth USD 6.5 billion. Had renewable generation been doubled, the avoided fuel cost would have reached USD 12.9 billion – more than twice the baseline saving.

EXECUTIVE SUMMARY

Renewable energy capacity additions reached a new high in 2025. Global renewable power capacity additions exceeded 690 gigawatts (GW) in 2025 – around one-fifth more than in 2024. Solar photovoltaics (PV) contributed more than 500 GW of this total, while wind added around 160 GW. Asia – and above all, China – remained the centre of deployment, with China accounting for close to two-thirds of new solar and three-quarters of new wind capacity. In 2025, renewable energy also overtook coal in terms of total, global installed capacity (IRENA, 2026a) to become the world's largest source of electricity.

Variable renewable costs fell as most dispatchable renewable energy costs rose. In 2025, the global, weighted-average levelised cost of electricity (LCOE) for solar PV remained unchanged compared to 2024, at USD 44 per megawatt hour (MWh).[3] Over the same period, the LCOE of onshore wind decreased to USD 33/MWh and offshore wind to USD 78/MWh. Driving these changes were two opposing factors that largely offset each other: on the one hand, cheaper equipment and lower installed costs; on the other, a higher cost of capital combined with slightly lower capacity factors.

The increases in the costs of most dispatchable renewables was driven by lower output, rather than changes in technology costs. Hydropower rose to USD 62/MWh, geothermal to USD 89/MWh and concentrated solar power (CSP) to USD 115/MWh. Bioenergy was the exception, with LCOE declining to USD 86/MWh due to increased output.

[3] *All values in 2025 USD.*

Figure S1 LCOE, total installed cost and capacity factor trends by renewable technology, 2024–2025*



		Onshore wind	Solar PV	Offshore wind	CSP	Hydropower	Geothermal	Bioenergy
LEVELISED COST OF ELECTRICITY	2025 (USD/MWh)	33	44	78	115	62	89	86
	Change 2024–2025	-4%	1%	-3%	9%	9%	53%	-2%
TOTAL INSTALLED COST	2025 (USD/kW)	976	667	2 931	2 418	2 079	5 997	3 606
	Change 2024–2025	-8%	-6%	0%	-36%	-10%	46%	9%
CAPACITY FACTOR	2025	36%	16%	41%	22%	43%	85%	78%
	Change 2024–2025**	2%	-1%	-1%	-19%	-5%	-3%	5%

Notes: * Figures are 2025 USD values; changes are compared to 2024 levels; ** all changes in capacity factor are expressed in percentage points; CF = capacity factor; CSP = concentrated solar power; kW = kilowatt; LCOE = levelised cost of electricity; PV = photovoltaic; TIC = total installed cost.

Renewables retained their cost advantage over fossil fuels. Despite the plateau, renewables remained the most cost-competitive source of new electricity in most markets in 2025. Their advantage over fossil fuels also continued to widen. Since 2010, the LCOE of solar PV has fallen by 89%, CSP by 72%, onshore wind by 71%, offshore wind by 63% and bioenergy by 3%. In contrast, the LCOEs of hydropower and geothermal increased by 41% and 53%, respectively.

The cost of new fossil fuel-fired generation continued to rise. A gas-turbine shortage, driven in part by surging data-centre demand, roughly doubled the capital expenditure associated with a new, combined-cycle plant. In the United States, this level rose to around USD 2 400/kW, while the LCOE from these plants in high gas price markets, such as Italy, Germany and Japan, rose towards USD 100/MWh. As in 2024, more than 90% of utility-scale renewable projects commissioned in 2025 were cheaper than the lowest-cost, new fossil-fuel alternative.

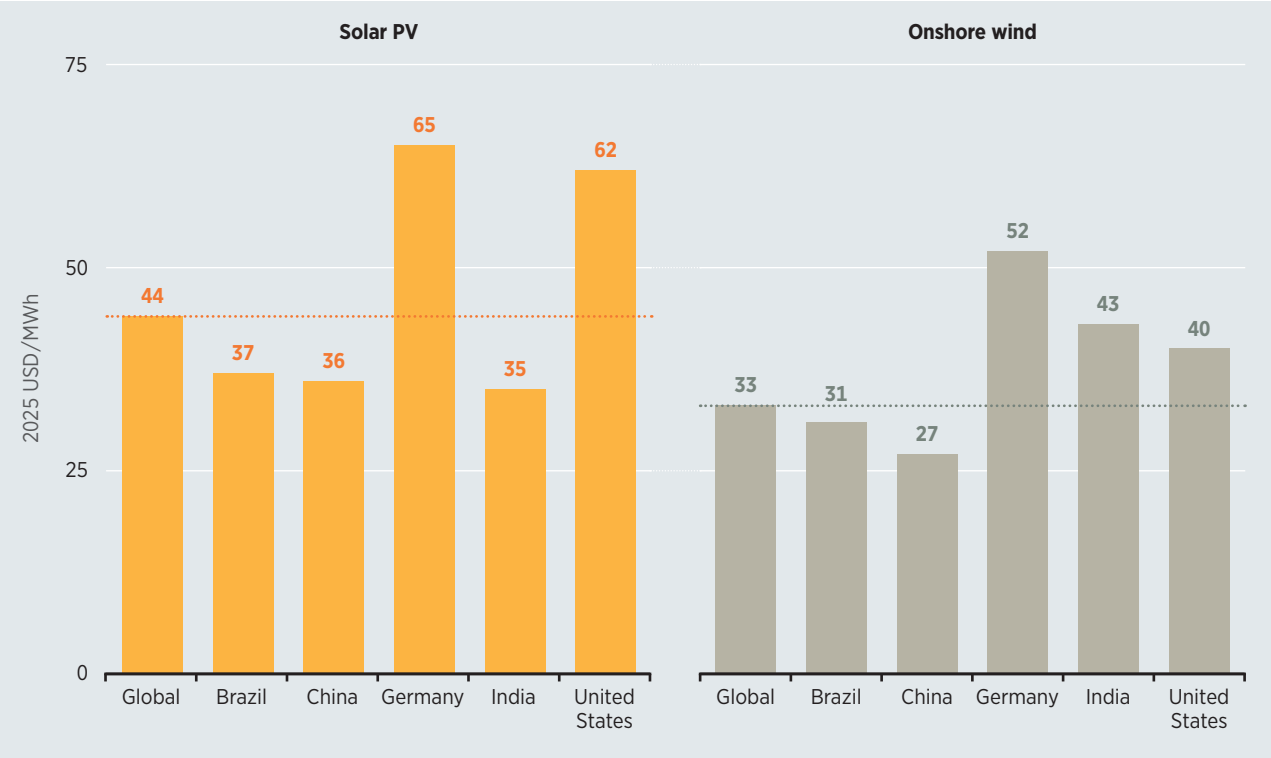
Figure S2 LCOE of renewable power technologies, 2010–2025 (USD/MWh)



Notes: CSP = concentrated solar power; MWh = megawatt hour; PV = photovoltaic; USD = United States dollar.

Cost trends diverged across technologies and regions. Stable global, cost-weighted averages in 2025 masked substantial differences across markets. China recorded the lowest onshore wind LCOE, at USD 27/MWh, and among the lowest solar PV LCOEs, at USD 36/MWh. Across major markets, the LCOE of solar PV ranged from USD 35/MWh in India to USD 65/MWh in Germany. LCOE values in the United States and Germany were also close to double those in China. This was due to permitting delays, interconnection bottlenecks and higher balance-of-system costs. For onshore wind, the LCOE ranged from USD 27/MWh in China to USD 52/MWh in Germany.

Figure S3 Solar PV and onshore wind LCOEs across major markets, 2025 (USD/MWh)



Notes: MWh = megawatt hour; PV = photovoltaic; USD = United States dollar.

Rapid declines in battery costs are transforming the economics of renewable power. In 2025, battery costs fell faster than those of any other energy technology. IRENA estimates the installed cost of a four-hour utility-scale battery at USD 140/kWh, a decrease of close to 30% in a single year and around 95% since 2010. As storage has become more affordable, solar and wind projects are increasingly being paired with battery energy storage systems. This is improving the utilisation of grid connections, shifting generation to periods of higher demand and reduce exposure to low or volatile electricity prices.



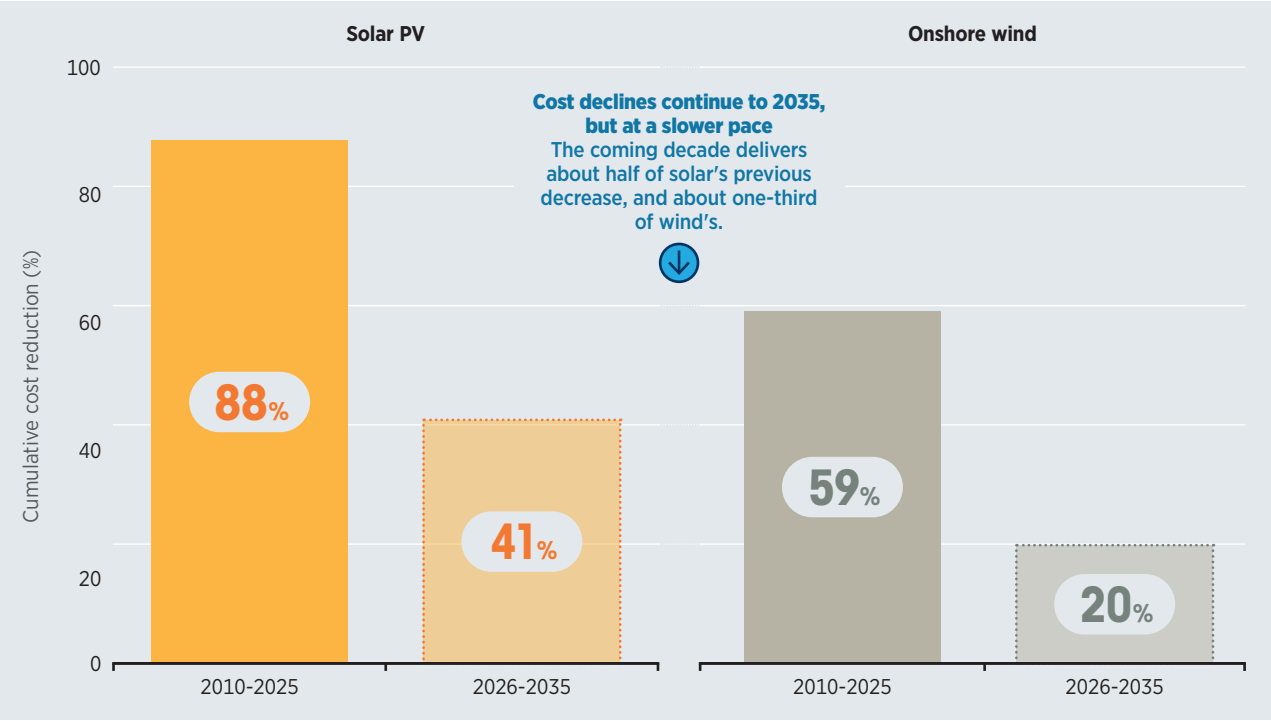
Cost reductions are expected to continue, but at a slower pace. After more than a decade of rapid, learning-driven cost reductions, solar PV and onshore wind have entered a phase of maturity. When permitting, grid connection, engineering and financing costs are included, between 2010 and around 2021, the costs of these two technologies fell, as experience accumulated and risk perceptions adjusted. The remarkable price decreases that followed – with solar module prices dropping by roughly two-thirds between 2022 and 2024 – were primarily driven by a substantial expansion of manufacturing capacity in China. This build-up resulted in a highly competitive landscape characterised by thin margins and prices approaching the cost of production.

This phase of intense competition may now be shifting. Clean-technology manufacturing investment more than halved between 2023 and the end of 2025. At the same time, China is actively consolidating its renewable energy manufacturing sector, while in April 2026 it also removed the value-added tax rebate on exports of certain solar PV products.

In parallel, commodity and component prices have been rising, pushing up Western onshore wind turbine prices, for example Looking ahead, these developments are likely to exert upward pressure on TICs throughout 2026.

Over the longer term, costs will continue to decline out to 2035, though far more slowly than before. Over the coming decade, TICs are projected to fall by around 40% for solar PV and 20% for onshore wind. These factors are roughly one-half and one-third, respectively, of the reductions recorded since 2010.

Figure S4 Cumulative reduction in global, weighted-average total installed costs for solar PV and onshore wind, 2010–2025 and 2026–2035 (%)



Note: PV = photovoltaic.

Cost reductions are expected to remain uneven across regions. China is set to remain the lowest-cost producer of both solar PV and onshore wind technologies, while the United States and Europe retain a persistent, though narrowing, cost premium. A comparable picture holds for energy storage, both short duration and long duration, with costs projected to continue to decline.

The figures given here are, however, model-based and indicative rather than predictive. Over a ten-year period, substantial uncertainty remains, as global geopolitical developments, industrial policy and supply-chain changes could result in costs being either higher or lower than these projections.

Cost of capital remains the decisive divide between markets. TICs have fallen in price worldwide, yet LCOEs still differ widely between markets; financing now explains much of that gap. IRENA data make the point directly: solar PV installed costs fell by around 6% in 2025, yet the LCOE of this technology remained unchanged. This was because a higher cost of capital offset the cheaper equipment and systems. The biggest cost differences have become driven more by where a project is built than by what is built.

IRENA's cost-of-capital model suggests that country-level macroeconomic conditions, such as sovereign risk, interest rates and inflation, explain about 56% of the variation in financing costs. At 24%, this is roughly 2.3 times the share attributable to technology. The same technology can therefore cost far more to finance in a high-risk economy than in a mature one. This impact is most significant for emerging and developing economies, where most new demand will arise. In these markets, the cost of capital – not the cost of technology – is the binding constraint, with targeted de-risking the critical lever.

Globally, the use of renewables avoided 8.4 gigatonnes of carbon dioxide (CO_2) emissions and saved USD 480 billion in fossil-fuel costs in 2025. Because renewable energy facilities require no fuel once operational, this saving is counter-cyclical: their value increases with rising global fossil-fuel prices – and they do so without any new investment, since the capacity is already in place.

Renewable energy is proving itself as a geopolitical shock absorber. In 2025, the renewable generation of three, import-exposed Southeast Asian economies – Indonesia, Thailand and the Philippines – displaced an estimated USD 5.7 billion of coal and gas that would otherwise have been bought, much of it imported.


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